Appetite 214 (2025) 108218

Contents lists available at ScienceDirect

Appetite

journal homepage: www.elsevier.com/locate/appet

AI-driven nudges: Influence of chatbot personas on consumer food choices

Xiyu Guo, Xiaoang Wan *

Department of Psychological and Cognitive Sciences, Tsinghua University, Beijing, China

ARTICLE INFO

Keywords:
Food choice
Meat reduction
Healthy eating
Human-AI interaction
Chatbot
Persona customization

ABSTRACT

Customized chatbot personas can meet the personalized needs of consumers and enhance user engagement in human-chatbot interactions. We conducted a study to investigate the influence of interacting with a chatbot customized as a health coaching assistant versus a gourmet recommendation assistant on participants' food choices. Participants were instructed to choose three dishes from a set of two meat and two vegetable options to compose a three-dish meal, both before and after they interacted with the health coaching assistant or the gourmet recommendation assistant. The results revealed a significant decrease in the frequency of meat-heavy meal selections after participants interacted with the health coaching assistant, compared to the gourmet recommendation assistant. These findings suggest that interacting with chatbots tailored as health coaching assistants can effectively nudge consumers toward more sustainable food choices. These findings highlight the potential of human-AI interactions in promoting sustainability and offer implications for the design and implementation of chatbots in the restaurant industry.

1. Introduction

Consumer needs and expectations have become increasingly diverse with the continuous evolution of information and communication technologies. For businesses, using these new technologies to provide products and services has emerged as a crucial strategy for enhancing competitiveness (Kao & Huang, 2023). In the digital age, artificial intelligence (AI) stands out as one of the most impactful technologies (Schwab, 2017). Therefore, restaurants are currently undergoing digital transformation, integrating AI technology into their operations to thrive in a highly competitive environment (Oğan, 2024). Such integration is likely to influence consumer behavior, particularly in food selection, through ways such as providing real-time information, offering personalized recommendations, and enhancing the dining experience, potentially promoting healthier eating.

Despite the potential benefits of robotics in restaurants, their widespread adoption may be hindered by the high costs associated with purchasing, leasing, and maintaining robotic systems (Berezina et al., 2019; Ivanov & Webster, 2018). AI-powered chatbots have emerged as a viable alternative to human agents to perform certain tasks in service delivery within the restaurant sector (Gupta et al., 2024; Leung & Wen, 2020). Chatbots are computer programs designed to simulate human-like conversations through voice or text-based interactions, and function as virtual assistants for users (Luo et al., 2019). Notable examples include DOM at Domino's Pizza and My Starbucks Barista at Starbucks, where chatbots are utilized to provide continuous and efficient customer service. The usage of AI-powered chatbots offers significant advantages such as reduced operational costs, enhanced service efficiency, and improved consumer experiences (Cheng et al., 2022).

That said, the impact of human-chatbot interactions on consumers' eating behaviors remains unclear. Existing literature on chatbots mainly focuses on technical aspects (Gupta et al., 2024; Kim et al., 2020) and consumers' attitudes and usage patterns regarding them (Liu et al., 2024; Osei et al., 2024; Yoon & Yu, 2022). Considering that dining has social, cultural, and psychological aspects (Pliner & Bell, 2009), it is important to explore the impact of human-chatbot interactions on consumers' eating behaviors. Understanding such impact can also help restaurants design better chatbot services to boost consumers' dining experience and loyalty.

Chatbots are revolutionizing how consumers interact with computers (Chaves & Gerosa, 2020). However, most existing large language model (LLM)-based chatbots, such as ChatGPT (Kalla et al., 2023), often remain generic and fail to adequately address users' personalized needs (Brandtzaeg & Følstad, 2018), resulting in unsatisfactory interactions (Gao et al., 2018, pp. 1371–1374). Consequently, the importance of tailoring personas of LLM-based conversational agents has gained increasing recognition (Ha et al., 2024, pp. 1–24). Previous research has highlighted the effectiveness of customized chatbot personas in

* Corresponding author. Department of Psychological and Cognitive Sciences, Tsinghua University, Beijing, 100084, China.
E-mail address: wanxa@mail.tsinghua.edu.cn (X. Wan).

https://doi.org/10.1016/j.appet.2025.108218
Received 24 March 2025; Received in revised form 1 July 2025; Accepted 2 July 2025
Available online 3 July 2025
0195-6663/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

facilitating more satisfying and human-like interactions (Moussawi & Benbunan-Fich, 2020). These studies demonstrate the value of designing chatbot personas by assigning distinct characteristics, behaviors, and contexts aligned with user preferences, thereby promoting more engaging and meaningful conversations (Ait Baha, et al., 2023). However, less attention has been given to how chatbot personas influence consumer choices, particularly in domains where decision-making is complex and context-dependent. In the present study, we investigated the impact of chatbots with different personas on the food choices participants made.

By examining the influence of divergent chatbot personas on consumers' food choices, we aimed to uncover the mechanisms through which these personas shape decision-making in food-related contexts. The findings of this research can provide a better understanding of the effects of human-AI interactions on consumers' eating behaviors and offer valuable insights into leveraging chatbots to promote sustainable food choices. Moreover, our findings can provide practical guidance for designing chatbots with specific personas in the restaurant industry, and help restaurants tailor their chatbot's persona to meet the needs and preferences of their target customers, ultimately enhancing the dining experience and customer loyalty.

2. Literature review and development of hypotheses

2.1. Chatbots in the restaurant industry

AI technology has permeated many industries, such as automotive, healthcare, education, and finance. Currently, it is making inroads into the restaurant industry (Chi et al., 2020). In the restaurant sector, AI-powered robots can serve a variety of functions both at the front and at the back of restaurants (Berezina et al., 2019). However, a few factors limit the usage of robots in the restaurant industry. For example, the acquisition, rental, and maintenance of robots may entail high costs (Ivanov & Webster, 2018). For another example, robots often necessitate ramps, special tracks, or platforms, which can impose constraints on the design and layout of a restaurant (Ivanov & Webster, 2017). In contrast, AI-powered chatbots may be more widely used in the restaurant industry, presumably as they offer a more affordable solution and round-the-clock availability (Sam & Jasim, 2023).

Chatbots were initially developed in the early 1960s to facilitate user interaction through human-like natural language (Ciechanowski et al., 2019). Users typically engage with chatbots through text-based interfaces or voice interactions (Rzepka et al., 2022). In recent years, significant advancements in natural language processing, machine learning, and deep learning have enhanced chatbots' ability to comprehend and engage in human-like conversations (Juraimi et al., 2024). These technological improvements have expanded the potential applications of chatbots in a variety of industries (Suhaili et al., 2021), especially within the restaurant sector (Alawami et al., 2025).

In the restaurant industry, chatbots are ushering in novel forms of interaction between consumers and brands (Paul et al., 2023). Restaurants can leverage chatbots to offer diverse services, such as promptly responding to messages, facilitating quick access to reservation systems or purchase options, handling takeout orders, and providing personalized recommendations (Leung & Wen, 2020; Yoon & Yu, 2022). Such technologies can enhance the efficiency of restaurant staff, enable seamless and effortless interactions between businesses and guests, enhance personalization by tailoring experiences, and even provide entertainment for guests (Ivanov & Webster, 2018).

2.2. Influence of chatbots on food choices

In the literature, many studies have identified a number of biological, psychological, economic, and environmental factors that can influence consumers' food choices (for comprehensive reviews see Symmank et al., 2017). Some of these factors are shown to act as barriers to achieving a healthy or sustainable diet. For example, the pursuit of hedonic pleasure can serve as a primary driver of food purchases (Mai & Hoffmann, 2015), which can conflict with motivations for healthy eating, ultimately leading individuals to choose less healthy foods (Briers et al., 2020). These findings highlight the importance of developing effective interventions to promote sustainable food choices (Cleland et al., 2025), yielding dual benefits for both consumers (Issock et al., 2023) and businesses (Tian & Kamran, 2021).

Recent studies have explored how to utilize chatbots to influence people's eating behaviors (Chan et al., 2022; Oh et al., 2021). Certain chatbots have been developed with the aim of guiding individuals toward healthy eating habits by providing tailored nutritional advice, setting achievable goals, and using persuasive messaging (Prasetyo et al., 2021, pp. 436–439; Rahmanti et al., 2022). These findings demonstrate that chatbots exert great potential in promoting healthy eating. Building on chatbots' ability to deliver information through natural dialogue (Wei et al., 2022), we posit that certain human-chatbot interactions may effectively influence user decision-making patterns.

2.3. Chatbot personas

When interacting with chatbots, users may hold varying expectations. However, many current chatbots fail to consider users' preferences and needs, which can result in biases and negative experiences for certain user groups (El Hefny et al., 2021, pp. 121–129). Therefore, the demand for personalized conversational experiences that cater to users' unique requirements is growing (Ha et al., 2024, pp. 1–24). Traditionally, personalized AI systems create customized experiences by inferring user preferences from usage data, a process known as data-driven personalization (Maksimova et al., 2021; Shumanov & Johnson, 2021). With the emergence of the GPT (Generative Pre-trained Transformer), users now have greater opportunities to actively engage in the customization process, a paradigm known as user-participated customization (Song et al., 2024, pp. 1–7).

Integrating personas into chatbot systems has been considered as a crucial component for enhancing personalization and user engagement in human-chatbot interactions (Wald et al., 2021, pp. 382–387). Customized chatbot personas can foster deeper emotional connections with users and promote trust via contextually appropriate communication styles (Lessio & Morris, 2020, pp. 3221–3228). Moreover, personas-driven chatbots can enhance service value by aligning interactions with user preferences and goals (Chaves & Gerosa, 2020). With the advancement of LLM technology, the potential for implementing more diverse and flexible characters in chatbots has significantly expanded (Deshpande et al., 2023). When making food choices, consumers are often guided by either hedonic or health-related goals (Bauer & Reisch, 2019; Elliston et al., 2017). Therefore, in the present study, we customized two chatbots with different personas, i.e., a health coaching assistant and a gourmet recommendation assistant, to explore the impact of interacting with these chatbots on participants' food choices.

Previous research has shown that chatbots customized with specific personas can engage users in a more relevant and tailored manner (Song et al., 2024, pp. 1–7). It should be noted that rich and contextually relevant information is often delivered during tailored human-chatbot interactions. For example, a health coaching assistant may emphasize health-related information, whereas a gourmet recommendation assistant may focus on pleasure-inducing content. Extensive studies have shown that health-related information can effectively nudge people toward healthier food choices (e.g., Bauer, van der Laan, de Bruijn, & Reisch, 2022; Carfora et al., 2019a; Carfora et al., 2019b; Wilson et al., 2016). One plausible explanation is that such information can serve as a priming to improve consumers' health perceptions and align their decisions with long-term wellness goals (Lähteenmäki, 2013; Papies, 2016). In contrast, pleasure-inducing information can activate short-term indulgence goals (Elliston et al., 2017; Harris et al., 2009),

which can override competing long-term goals, such as maintaining a healthy diet (El Dahr & Fort, 2008).

Moreover, according to social influence theory and the elaboration likelihood model, persuasive communication can lead to changes in attitudes, which can subsequently influence behavior (Petty & Cacioppo, 1986; Wood, 2000). Previous studies have highlighted the role of information interventions in shaping attitudes and behaviors related to meat consumption (Bertolotti et al., 2016; Carfora et al., 2019a; Carfora et al., 2019b). In particular, health-focused messaging has been shown to reduce individuals' inclination toward meat consumption (de Boer et al., 2017). Based on these findings, we hypothesized that participants would choose meat-heavy meals less frequently after they interacted with a chatbot customized as a health coaching assistant, compared to one tailored as a gourmet recommendation assistant.

2.4. The present study

In the present study, we examined how interacting with chatbots customized to different personas might influence food choices. Participants completed a food choice task before and after interacting with either a chatbot customized as a health coaching assistant or one tailored as a gourmet recommendation assistant. Food choices were measured using a "three-out-of-four" task (Guo et al., 2024; Wan et al., 2022; Wang & Wan, 2025), designed to reflect typical Chinese meal compositions, which typically include a staple food (e.g., rice) and mixed dishes categorized as meat- or vegetable-based (Zhai et al., 2014). Specifically, participants were asked to choose three dishes from a set of two meat dishes and two vegetable dishes, resulting in either a meat-heavy meal (two meat dishes and one vegetable dish) or a vegetable-forward meal (two vegetable dishes and one meat dish). Both options provide a balanced diet of vegetables and meat, which is essential for promoting health, reducing hunger, and improving survival (Muehlhoff et al., 2013). However, encouraging fewer meat-heavy meals may result in meat reduction (Qiu & Wan, 2024).

3. Methods

3.1. Samples and participants

We used the G*Power software to conduct a sample size estimation (Faul et al., 2007). For the 2 × 2 × 2 mixed-design being used in this study, the minimum sample size was 56 for a medium effect size f of 0.25, with a statistical power of 0.85 and an alpha of 0.05. A total of sixty young Chinese adults (mean age = 25.2 ± 2.62 years, ranging from 19 to 30 years; 40 females and 20 males) with normal body weights (mean BMI = 20.5 ± 2.12 kg/m^2) were recruited to participate in this study via the Credamo platform (https://www.credamo.com).

In the present study, all participants reported that they had no dietary restrictions, had not recently been on a diet or weight loss program, and had normal or corrected-to-normal vision without color blindness. All participants were also asked to refrain from eating for at least 1 h prior to this study. This study was performed in accordance with the Declaration of Helsinki and was approved by the Institutional Review Board of a major research university in Beijing, China. This work was preregistered at the Open Science Framework (https://osf.io/gtekz). Each participant signed an informed consent electronically and received 9 Chinese yuan (equal to approximately 1.24 USD) after they had completed the study.

3.2. Chatbot design

We developed two chatbots with different personas, i.e., a health coaching assistant and a gourmet recommendation assistant, using the CharacterGLM Pro model on the AI Utopia PRO platform (https://pro.aitopia.com). This platform enables the automatic generation of personalized AI characters by inputting background information, allowing users to engage in realistic interactions with customized AI personas (Zhou et al., 2024, pp. 1457–1476). As shown in Table 1, we defined each persona through distinct operational prompts input to the AI Utopia PRO platform, which autonomously generated the corresponding conversational styles. The health coaching persona was designed to promote nutritious eating through professional yet warm

Table 1
Chatbot persona definitions: Operational prompts used for AI configuration and corresponding participant-facing descriptions.

	Persona	
AI Configuration Prompt	Health coaching	You are designed to promote healthy diets and lifestyles. Your goal is to help users improve their well-being by sharing insights on nutritious foods, dietary knowledge, and healthy living tips. You are professional, warm, and positive, offering authoritative health advice while expressing genuine concern for users' health. You advocate for a healthy lifestyle and communicates expert knowledge in an accessible manner, using positive and encouraging language to provide specific, actionable suggestions.
	Gourmet recommendation	You focus on the enjoyment of delicious food and the dining pleasure. Your mission is to enrich users' lives by sharing delectable food ideas, unique dishes, and dining tips. You are enthusiastic and passionate about flavorful food, emphasizing the sensory pleasure it brings. You aim to inspire users' passion for gourmet experiences by recommending mouth-watering and special dishes. Your language style is sensual and captivating, emphasizing the satisfaction, happiness, and enjoyment that gourmet food provides, while sparking users' appetites and culinary curiosity.
Participant-Facing Description	Health coaching	Qingqing is a health coaching assistant who focuses on healthy diet and lifestyle. Qingqing is enthusiastic, cheerful, responsible, and has a great passion for health and nutrition. Qingqing believes that "health is the cornerstone of a happy life," and is committed to helping people improve their health through a scientific diet and lifestyle. Qingqing can help more people improve their well-being brought by health by sharing insights on nutritious food, dietary knowledge, and healthy living tips.
	Gourmet recommendation	Qingqing is a gourmet recommendation assistant that focuses on the enjoyment of delicious food and the dining pleasure. Qingqing is enthusiastic, curious, and has a great passion for gourmet experiences. Qingqing believes that "gourmet is the greatest enjoyment in life," and is committed to bringing people happiness and satisfaction through gourmet and dining experience. Qingqing can help more people enrich their lives brought by gourmet by recommending mouth-watering food, unique dishes, and dining tips.

Note: Both the operational prompts and persona descriptions were originally implemented in Chinese and have been translated into English for publication.

guidance, whereas the gourmet persona focused on enhancing culinary enjoyment through passionate food recommendations. To ensure clarity for participants, both chatbots, named Qingqing, were introduced with simplified descriptions matching their respective personas (also see Table 1).

3.3. Food choice task

For the food choice task, we selected six images of Chinese dishes similar to those used by Guo et al. (2024). Half of these images presented meat-heavy dishes (i.e., beef brisket with black pepper, braised ribs in sauce, and braised chicken in sauce), and the other half presented vegetable-forward dishes (i.e., garlic minced cabbage, stir-fried leaf lettuce, and hand-torn cabbage). All dishes were served in white ceramic containers. When each food image (subtending 150 × 150 pixels) was shown on the screen, a dish name consisting of four Chinese characters also appeared below the image to ensure understanding (see Fig. 1 for an illustration).

The food choice task consisted of six trials. Within each trial, participants were instructed to choose three dishes as a meal from two meat and two vegetable dishes arranged in a 2 × 2 matrix format on the screen (see Fig. 1). Dish names were displayed beneath corresponding images, with participants indicating their selections by checking boxes adjacent to chosen items. We used six different choice sets, with the placement of meat or vegetable dishes counterbalanced across trials to minimize position-based biases. The six choice sets were presented in a random order to ensure experimental control. To measure participants' food choices, we calculated the percentages of the meat-heavy meals they chose.

3.4. Measures

In this study, we also asked our participants to complete a series of ratings on the chatbot they interacted with. First, participants were required to rate the extent to which the information provided by the chatbot during the interaction was related to health or pleasure, and the extent to which the chatbot was like a human or a machine (Mende et al., 2019). All ratings are on 7-point Likert scales, with greater mean scores indicating greater intensity of the attributes being rated.

Second, participants were asked to rate their attitudes toward the chatbot and their intention to use the chatbot via the following scales. For the measurement of attitudes, we used a three-item 7-point Likert scale used by Akdim et al. (2023). Higher mean scores indicate that participants show more positive attitudes toward the chatbot. For the measurement of intention to use, we used a four-item 7-point Likert scale used by Piao et al. (2020), with greater mean scores indicating greater intention to use the chatbot.

Given that people differ in the extent to which they incorporate taste and health motives into their food choices, we also measured the importance that participants attached to perceived health and pleasure features of food choices using the General Health Interest subscale and the Pleasure subscale of the Health and Taste Attitudes toward Foods scale (Roininen et al., 1999), respectively. The General Health Interest subscale consists of eight statements, and the Pleasure subscale consists of six statements. Both subscales are 7-point scales, with greater mean scores indicating a higher level of health or hedonic motivation when participants made food choices.

3.5. Design and procedure

Considering documented sex-related differences in food choices (e.g., Dibb-Smith & Brindal, 2015), we used a 2 (Chatbot Persona: health coaching assistant or gourmet recommendation assistant) × 2 (Stage: pre- or post-interaction) × 2 (Participant Sex: male or female) mixed design. Note that Chatbot Persona and Participant Sex were treated as between-participant factors, whereas Stage was treated as a within-participant factor. That is, participants were randomly divided into two groups. One group of participants was asked to interact with the chatbot customized as a health coaching assistant (i.e., health group), whereas the other group was asked to interact with the chatbot customized as a gourmet recommendation assistant (i.e., gourmet group).

The main experiment consisted of four stages. First, participants were asked to imagine planning to eat at a new restaurant and, in order to prepare in advance, they needed to decide what they would eat using the food choice task mentioned above. Second, each participant was asked to imagine that his or her was waiting in line for a table at the new restaurant. During the waiting period, the restaurant introduced a chatbot named Qingqing, which was designed to interact with consumers and provide companionship. Participants were asked to interact with the chatbot. Specifically, participants in the health group were instructed to share their healthy eating efforts, challenges, or successes

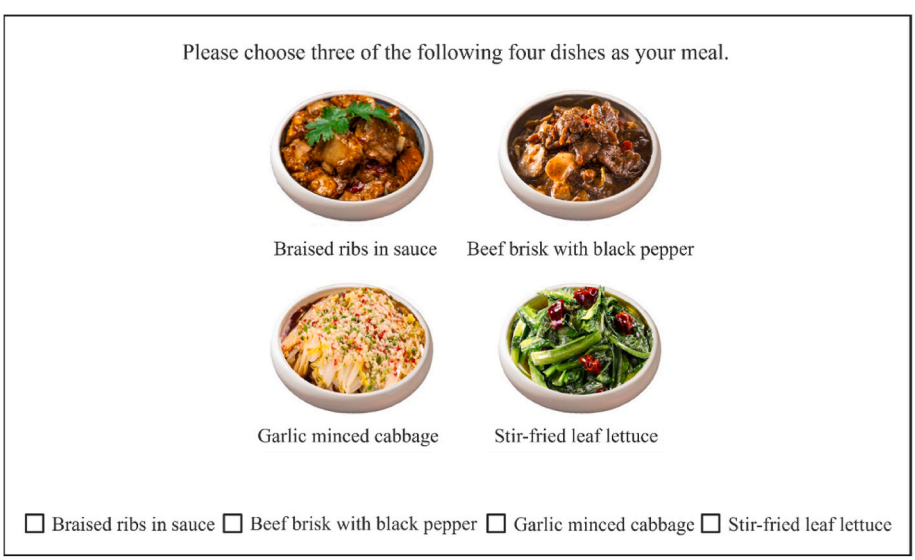

Fig. 1. The food selection interface used in this study. Participants were instructed to choose any three dishes from four options (two meat and two vegetable dishes) arranged in a 2 × 2 grid. Each dish was presented with its name, and selections were made using interactive checkboxes below the grid. For illustration purposes, original Chinese dish names have been translated to English.

with the chatbot, whereas those in the gourmet group were encouraged to discuss their taste preferences, culinary experiences, or favorite dishes. Before the interactions, we presented participants with a brief description to help them better understand the chatbot (see Table 1). Third, after the interactions, participants were asked to imagine dining at the restaurant and complete the food choice task again to select their meal. Fourth, participants rated the chatbot's information relevance and human-likeness, while also reporting their personal attitudes toward and usage intentions regarding the chatbot, and indicating their health or hedonic motivation in food choices. Moreover, participants were asked to indicate their frequency of using chatbots in their daily life (never, occasionally, sometimes, or often).

At the end of the study, participants were also asked to rate the palatability, pleasantness, energy-density, and healthiness of, as well as their familiarity with, each dish. These ratings were completed on 7-point scales, with larger numbers indicating a higher level of intensity of the attributes being rated. The entire experiment lasted approximately 15 min.

4. Results

4.1. Food choices

We calculated the mean choice percentages of the meat-heavy meals under different conditions in this study. Then we performed a 2 (Chatbot Persona: health coaching assistant or gourmet recommendation assistant) × 2 (Stage: pre- or post-interaction) × 2 (Participant Sex: male or female) mixed-design analysis of variance (ANOVA) on the data. As summarized in Table 2, the results revealed a significant main effect of Stage, and a significant interaction term between Chatbot Persona and Stage. None of other main or interaction effects was significant.

To interpret the interaction term, we conducted simple-effect analyses. As shown in Fig. 2, the results showed comparable choice percentages of meat-heavy meals before participants interacted with the health coaching assistant (M = 81.1 %, SD = 32.1 %) or the gourmet recommendation assistant (M = 68.9 %, SD = 37.8 %), $t(58) = 1.35$, $p = 0.182$, Cohen's $d = 0.35$. However, participants in the health group chose meat-heavy meals less frequently (M = 40.0 %, SD = 43.7 %) than their counterparts in the gourmet group (M = 68.9 %, SD = 38.3 %), $t(58) = 2.72$, $p = 0.009$, Cohen's $d = 0.70$.

We also calculated the mean scores of participants' health or hedonic motivation, and used them as the covariates in the two Chatbot Persona × Stage × Participant Sex analyses of covariance (ANCOVAs) we performed on the food choice data, respectively. When controlling for health motivation as a covariate, the results revealed a significant interaction term on the choice percentages of the meat-heavy meals, $F(1, 55) = 20.02$, $p < 0.001$, $\eta_p^2 = 0.27$. A similar pattern emerged when hedonic motivation was used as a covariate, with the interaction term remaining statistically significant, $F(1, 55) = 19.09$, $p < 0.001$, $\eta_p^2 = 0.26$. These results indicated that participants' health or hedonic motivations did not modulate the effects of chatbot personas on their food choices.

Moreover, we calculated the time participants in both groups spent interacting with their respective chatbots. Independent-samples t-tests revealed comparable interaction durations between the health coaching assistant (M = 208 s, SD = 109 s) and the gourmet recommendation assistant (M = 188 s, SD = 108 s), $t(58) = 0.70$, $p = 0.486$, Cohen's $d = 0.18$. When interaction time was included as a covariate in the Chatbot Persona × Stage × Participant Sex ANCOVA for food choice data, it showed no significant moderating effect. The analysis instead showed a significant Chatbot Persona × Stage interaction, $F(1, 55) = 20.89$, $p < 0.001$, $\eta_p^2 = 0.28$, confirming that persona effects on food choices were independent of interaction duration.

Table 2
The results of ANOVA on the food choices in this study.

Effect	$F(1, 56)$	p	η_p^2
Chatbot Persona	0.70	0.406	0.01
Stage	20.69	< 0.001	0.27
Participant Sex	0.73	0.396	0.01
Chatbot Persona × Stage	20.69	< 0.001	0.27
Chatbot Persona × Participant Sex	0.06	0.802	0.001
Stage × Participant Sex	0.03	0.859	0.001
Three-way interaction	0.03	0.859	0.001

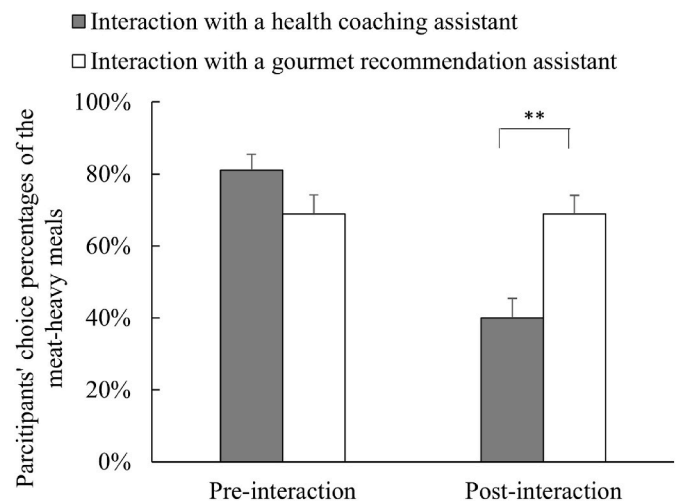

Fig. 2. Mean selection percentages of meat-heavy meals across experimental stages. Results show pre-interaction (left) and post-interaction (right) choices for participants in the health coaching condition (gray bars) versus gourmet recommendation condition (white bars). Note that the error bars show the standard errors of the means, and ** denotes $p < 0.01$.

4.2. Topic modeling for interactive content

To further analyze both chatbot personas, we applied Latent Dirichlet Allocation (LDA) to model topics within the interactive content (Jelodar et al., 2019). The results showed that interactions with the health coaching assistant featured topic words such as "food," "vegetables," "fruits," "nutrition," "eggs," and "protein," all related to health. In contrast, interactions with the gourmet recommendation assistant included topic words such as "experience," "try," "taste," "flavor," "mouthfeel," and "delicious," all associated with pleasure.

4.3. Ratings of chatbots

First, the mean scores of participants' perceptions of information relevance were calculated for both chatbots, followed by independent samples t-tests. The results revealed that the information provided by the health coaching assistant was more relevant to health ($M = 6.10$, $SD = 0.89$) than those provided by the gourmet recommendation assistant ($M = 3.07$, $SD = 1.72$), $t(58) = 8.59$, $p < 0.001$, Cohen's $d = 2.21$. Conversely, the information provided by the gourmet recommendation assistant was more relevant to pleasure ($M = 4.83$, $SD = 1.78$) than those provided by the health coaching assistant ($M = 2.13$, $SD = 0.86$), $t(58) = 7.47$, $p < 0.001$, Cohen's $d = 1.93$. The results indicated that our manipulation of both chatbot personas was valid.

Second, we calculated the mean scores of human-likeness for both chatbots. An independent samples t-test revealed no significant difference between the health coaching assistant ($M = 4.63$, $SD = 1.63$) and the gourmet recommendation assistant ($M = 4.62$, $SD = 1.82$), $t(58) = 0.04$, $p = 0.970$, Cohen's $d = 0.01$.

Third, mean scores of participants' attitudes toward the chatbots and their usage intentions were analyzed using independent samples t-tests. No significant differences emerged between the health coaching

assistant ($M = 5.17$, $SD = 1.48$) and the gourmet recommendation assistant ($M = 5.10$, $SD = 1.74$) in terms of attitude, $t(58) = 0.16$, $p = 0.873$, Cohen's $d = 0.04$. Similarly, no significant differences were observed between the health coaching assistant ($M = 5.08$, $SD = 1.48$) and the gourmet recommendation assistant ($M = 4.78$, $SD = 1.75$) in terms of usage intention, $t(58) = 0.74$, $p = 0.465$, Cohen's $d = 0.19$. Moreover, 28 %, 44 %, and 28 % of participants reported using chatbots occasionally, sometimes and often, respectively.

4.4. Ratings of food

We calculated the mean rating scores of the dishes, and performed paired samples *t*-tests on these data. As summarized in Table 3, the results revealed that the meat dishes were rated as more palatable, more pleasant, more energy-dense, and less healthy than the vegetable dishes were, but with comparable familiarity scores.

5. Discussion

In summary, our results revealed that participants selected significantly fewer meat-heavy meals after interacting with the health coaching assistant compared to the gourmet recommendation assistant. This effect persisted regardless of participants' baseline health or hedonic motivations, and remained unaffected by interaction duration (averaging just 3–3.5 min). These findings collectively demonstrate the health coaching assistant's superior efficacy in promoting sustainable food choices, independent of both intrinsic motivational factors and brief interaction time. The impact of chatbot personas may be attributed to the health-related information provided by the health coaching assistant, which may prime individuals to improve their health perceptions and motivations (Lähteenmäki, 2013; Papies, 2016). Alternatively, drawing on the social influence theory (Wood, 2000) and the elaboration likelihood model (Petty & Cacioppo, 1986), interacting with the health coaching assistant may shift attitudes toward healthy eating (de Boer et al., 2017), thus influencing food choices. In contrast, the gourmet recommendation assistant may emphasize pleasure-inducing information, activating short-term indulgence goals (Elliston et al., 2017; Harris et al., 2009) and potentially overriding long-term goals such as healthy eating (El Dahr & Fort, 2008). Thus, participants were more likely to make more sustainable food choices after interacting with health coaching assistants than with gourmet recommendation assistants.

This study can advance the literature in two key ways. First, our findings highlight the role of human-AI interactions in prompting sustainable food consumption. Although prior research has identified various behavioral interventions for sustainable consumption (e.g., Guo et al., 2024; Guo & Wan, 2025; Wang & Wan, 2025), our work allowed individuals to interact directly with chatbots. Our findings demonstrate that interactions with health coaching assistant can effectively promote more sustainable food selections. This finding not only extends the application of social influence theory (Wood, 2000) and the elaboration likelihood model (Petty & Cacioppo, 1986) to AI-driven health interventions, but also offers theoretical support for AI-based health tool design.

Second, the findings of this study may also contribute to human-AI interaction research. Given the progress in AI and the increasing use of AI techniques in human-oriented applications, the demand for personalized conversational experiences that meet users' unique requirements is increasing (Ha et al., 2024, pp. 1–24). Different from traditional data-driven personalized interactions based on large language models (Maksimova et al., 2021; Shumanov & Johnson, 2021), this study demonstrates the value of persona design in human-AI interactions through customized chatbot personas, such as the health coaching assistant or the gourmet recommendation assistant. These customized personas address individual user needs while enhancing the predictability and acceptance of chatbot behavior (Ha et al., 2024, pp. 1–24), providing a new direction for research and showing how persona-driven interactions can positively influence consumer decision-making.

This study also offers several practical implications. First, our findings demonstrate that even brief interactions with health-focused chatbots can significantly reduce meat-heavy meal selections, providing restaurants with a scalable tool to encourage sustainable choices without triggering defensive reactions (Zeng et al., 2024). Second, implementing customized chatbots allows restaurants to concurrently achieve public health goals and business objectives, enhancing customer satisfaction and brand reputation through behavior change support (Fadhil & Gabrielli, 2017, pp. 261–265; Tian & Kamran, 2021). Considering that significant effects emerged from brief interactions (3–3.5 min) in this study, longitudinal studies should be conducted to assess sustained impacts and further validated in real restaurant settings. Third, although focused on dietary choices, our persona-based framework shows transfer potential to other domains, including financial planning (e.g. chatbots promoting long-term saving) and mental health (e.g., sleep habit improvement).

As with any study, there are several limitations to the generalizability and interpretability of the present study. First, while our study examined individual food choices, real-world decisions often occur in social dining contexts (e.g., Cruwys et al., 2015; Wang et al., 2021), warranting future research on the effect of chatbot personas during commensality. Second, our study focused on text-based interactions, whereas voice-based interactions have demonstrated advantages in natural engagement and anthropomorphism (Liebrecht et al., 2023; Yang & Lee, 2019). Future research should examine how these modalities differentially influence dietary choices. Third, the generalizability of our findings may be limited by the exclusive use of young Chinese adults and culturally-specific food stimuli. Future cross-cultural replications should employ locally adapted food options while examining potential variations in chatbot effectiveness across different demographic groups, including age ranges, cultural backgrounds, and gender distributions (Folk et al., 2025; Jin & Eastin, 2023; van der Goot & Pilgrim, 2020).

6. Conclusion

In conclusion, this study demonstrates that interactions with persona-specific chatbot can effectively influence consumers' food choices. Our findings reveal that interacting with chatbots customized to be health coaching assistants, compared to gourmet recommendation assistants, can significantly reduce selections of meat-heavy choices.

Table 3
The mean scores (with SDs in parentheses) of the dishes, and the results of paired samples t-tests on these data in this study.

Dish		Palatability	Pleasantness	Energy-density	Healthiness	Familiarity
Meat	*M*	6.14	6.07	5.49	4.38	5.85
	(SD)	(0.56)	(0.60)	(0.73)	(1.07)	(0.73)
Vegetable	*M*	4.89	4.83	2.19	5.89	5.79
	(SD)	(0.94)	(0.94)	(0.73)	(0.63)	(0.73)
	t(59)	8.81	8.54	24.98	10.46	0.48
	p	< 0.001	< 0.001	< 0.001	< 0.001	0.632
	Cohen's *d*	1.14	1.10	3.22	1.35	0.06

These findings advance our understanding of how conversational AI can promote sustainable dietary behaviors, and provide practical insights for designing effective chatbot interventions. As AI becomes increasingly prevalent in consumer contexts, these findings offer valuable guidelines for developing chatbot personas that can support both public health and environmental sustainability goals.

CRediT authorship contribution statement

Xiyu Guo: Writing – original draft, Investigation. **Xiaoang Wan:** Writing – review & editing, Conceptualization.

Conflict of interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical statement

Ethical approval for the involvement of human subjects in this study was granted by Institution Review Board of Tsinghua University. This study was performed in accordance with the ethical standards laid down in the Declaration of Helsinki. Participants gave informed consent electronically via the statement "I am aware that my responses are confidential, and I agree to participate in this survey" where an affirmative reply was required to enter the survey. They were able to withdraw from the survey at any time without giving a reason. Participants were recruited and completed the experiment on the Credamo platform (www.credamo.com). All data were de-identified and only reported in the aggregate. All participants acknowledged an informed consent statement in order to participate in the study. They received monetary compensation after completion.

Funding Sources

This work was supported by the National Natural Science Foundation of China (Grant No. 72372081) awarded to Xiaoang Wan.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xiaoang Wan reports financial supports were provided by the National Natural Science Foundation of China.

Acknowledgements

The authors would like to thank Dr. Zhuang Chen and the AI Utopia Pro platform (https://pro.ai-topia.com) for their technical support.

Data availability

The data that supports the findings of this study are available from https://osf.io/gnv4q/.

References

Ait Baha, T., El Hajji, M., Es-Saady, Y., & Fadili, H. (2023). The power of personalization: A systematic review of personality-adaptive chatbots. *SN Computer Science, 4*(5), 661. https://doi.org/10.1007/s42979-023-02092-6

Akdim, K., Belanche, D., & Flavián, M. (2023). Attitudes toward service robots: Analyses of explicit and implicit attitudes based on anthropomorphism and construal level theory. *International Journal of Contemporary Hospitality Management, 35*(8), 2816–2837. https://doi.org/10.1108/IJCHM-12-2020-1406

Alawami, A., Alawami, M., Obaid, A., Al-rushaydan, M., Boudjedra, M. L., Boudjedra, B. E., Alharbi, M., Alzoori, H., Ab-osaif, S., Bshir, G., Alawami, H., & Abufawr, M. (2025). A systematic review of AI-driven innovations in the hospitality sector: Implications on restaurant management. *American Journal of Industrial and Business Management, 15*, 30–40. https://doi.org/10.4236/ajibm.2025.151003

Bauer, J. M., & Reisch, L. A. (2019). Behavioural insights and (un)healthy dietary choices: A review of current evidence. *Journal of Consumer Policy, 42*(1), 3–45. https://doi.org/10.1007/s10603-018-9387-y

Bauer, J. M., van der Laan, L. N., de Bruijn, G.-J., & Reisch, L. A. (2022). Battle of the primes – The effect and interplay of health and hedonic primes on food choice. *Appetite, 172*, Article 105956. https://doi.org/10.1016/j.appet.2022.105956

Berezina, K., Ciftci, O., & Cobanoglu, C. (2019). Robots, artificial intelligence, and service automation in restaurants. In S. Ivanov, & C. Webster (Eds.), *Robots, artificial intelligence, and service automation in travel, tourism and hospitality* (pp. 185–219). Emerald Publishing Limited. https://doi.org/10.1108/978-1-78756-687-320191010.

Bertolotti, M., Chirchiglia, G., & Catellani, P. (2016). Promoting change in meat consumption among the elderly: Factual and prefactual framing of health and well-being. *Appetite, 106*, 37–47. https://doi.org/10.1016/j.appet.2016.02.150

Brandtzaeg, P. B., & Følstad, A. (2018). Chatbots: Changing user needs and motivations. *Interactions, 25*(5), 38–43. https://doi.org/10.1145/3236669

Briers, B., Huh, Y. E., Chan, E., & Mukhopadhyay, A. (2020). The unhealthy = tasty belief is associated with BMI through reduced consumption of vegetables: A cross-national and mediational analysis. *Appetite, 150*, Article 104639. https://doi.org/10.1016/j.appet.2020.104639

Carfora, V., Bertolotti, M., & Catellani, P. (2019a). Informational and emotional daily messages to reduce red and processed meat consumption. *Appetite, 141*, Article 104331. https://doi.org/10.1016/j.appet.2019.104331

Carfora, V., Catellani, P., Caso, D., & Conner, M. (2019b). How to reduce red and processed meat consumption by daily text messages targeting environment or health benefits. *Journal of Environmental Psychology, 65*, Article 101319. https://doi.org/10.1016/j.jenvp.2019.101319

Chan, W., Fitzsimmons-Craft, E., Smith, A., Firebaugh, M., Fowler, L., DePietro, B., Topooco, N., Wilfley, D., Taylor, C., & Jacobson, N. (2022). The challenges in designing a prevention chatbot for eating disorders: Observational study. *JMIR Formative Research, 6*(1), Article e28003. https://doi.org/10.2196/28003

Chaves, A. P., & Gerosa, M. A. (2020). How should my chatbot interact? A survey on social characteristics in human–chatbot interaction design. *International Journal of Human-Computer Interaction, 37*(8), 729–758. https://doi.org/10.1080/10447318.2020.1841438

Cheng, X., Zhang, X., Cohen, J., & Mou, J. (2022). Human vs. AI: Understanding the impact of anthropomorphism on consumer response to chatbots from the perspective of trust and relationship norms. *Information Processing & Management, 59*(3), Article 102940. https://doi.org/10.1016/j.ipm.2022.102940

Chi, O. H., Denton, G., & Gursoy, D. (2020). Artificially intelligent device use in service delivery: A systematic review, synthesis, and research agenda. *Journal of Hospitality Marketing & Management, 29*(7), 757–786. https://doi.org/10.1080/19368623.2020.1721394

Ciechanowski, L., Przegalinska, A., Magnuski, M., & Gloor, P. (2019). In the shades of the uncanny valley: An experimental study of human–chatbot interaction. *Future Generation Computer Systems, 92*, 539–548. https://doi.org/10.1016/j.future.2018.01.055

Cleland, E., McBey, D., Darlene, V., McCormick, B. J. J., & Macdiarmid, J. I. (2025). Still eating like there's no tomorrow? A qualitative study to revisit attitudes and awareness around sustainable diets after 10 years. *Appetite, 206*, Article 107799. https://doi.org/10.1016/j.appet.2024.107799

Cruwys, T., Bevelander, K. E., & Hermans, R. C. (2015). Social modeling of eating: A review of when and why social influence affects food intake and choice. *Appetite, 86*, 3–18. https://doi.org/10.1016/j.appet.2014.08.035

de Boer, J., Schösler, H., & Aiking, H. (2017). Towards a reduced meat diet: Mindset and motivation of young vegetarians, low, medium and high meat-eaters. *Appetite, 113*, 387–397. https://doi.org/10.1016/j.appet.2017.03.007

Deshpande, A., Murahari, V., Rajpurohit, T., Kalyan, A., & Narasimhan, K. (2023). Toxicity in ChatGPT: Analyzing persona-assigned language models. *Findings of the association for computational linguistics: Emnlp 2023* (pp. 1236–1270). Singapore: Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.findings-emnlp.88

Dibb-Smith, A., & Brindal, E. (2015). Table for two: The effects of familiarity, sex and gender on food choice in imaginary dining scenarios. *Appetite, 95*, 492–499. https://doi.org/10.1016/j.appet.2015.07.032

El Dahr, H., & Fort, F. (2008). Effect of goal salience and goal conflict on typicality: The case of "health food". *Advances in Consumer Research, 35*, 282–288.

El Hefny, W., El Bolock, A., Herbert, C., & Abdennadher, S. (2021). Applying the character-based chatbots generation framework in education and healthcare. *Proceedings of the 9th international conference on human-agent interaction*. New York, NY: ACM. https://doi.org/10.1145/3472307.3484172

Elliston, K. G., Ferguson, S. G., Schüz, N., & Schüz, B. (2017). Situational cues and momentary food environment predict everyday eating behavior in adults with overweight and obesity. *Health Psychology, 36*(4), 337–345. https://doi.org/10.1037/hea0000439

Fadhil, A., & Gabrielli, S. (2017). Addressing challenges in promoting healthy lifestyles: The al-chatbot approach. *Proceedings of the 11th EAI international conference on pervasive computing technologies for healthcare*. New York, NY: ACM. https://doi.org/10.1145/3154862.3154914

Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods, 39*, 175–191. https://doi.org/10.3758/BF03193146

Folk, D. P., Wu, C., & Heine, S. J. (2025). Cultural variation in attitudes toward social chatbots. *Journal of Cross-Cultural Psychology, 56*(3), 219–239. https://doi.org/10.1177/00220221251317950

Gao, J., Galley, M., & Li, L. (2018). Neural approaches to conversational AI. *The 41st international ACM SIGIR conference on research & development in information retrieval*. New York, NY: ACM. https://doi.org/10.1145/3209978.3210183

Guo, X., & Wan, X. (2025). The robotic companion's choice encourages sustainable food choices: The moderating role of anthropomorphism. *Applied Psychology: Health and Well-Being, 17*(2), Article e70015. https://doi.org/10.1111/aphw.70015

Guo, X., Wang, C., & Wan, X. (2024). Influence of a robotic companion on women's food choices: Evidence from an imaginary task. *Applied Psychology: Health and Well-being, 16*(4), 2266–2282. https://doi.org/10.1111/aphw.12588

Gupta, M., Dheekonda, V., & Masum, M. (2024). Genie: Enhancing information management in the restaurant industry through AI-powered chatbot. *International Journal of Information Management Data Insights, 4*(2), Article 100255. https://doi.org/10.1016/j.jjimei.2024.100255

Ha, J., Jeon, H., Han, D., Seo, J., & Oh, C. (2024). CloChat: Understanding how people customize, interact, and experience personas in large language models. *Proceedings of the 2024 CHI conference on human factors in computing systems*. New York, NY: ACM. https://doi.org/10.1145/3613904.3642472

Harris, J. L., Bargh, J. A., & Brownell, K. D. (2009). Priming effects of television food advertising on eating behavior. *Health Psychology, 28*(4), 404–413. https://doi.org/10.1037/a0014399

Issock, P. B. I., Mpinganjira, M., & Roberts-Lombard, M. (2023). Beyond sustainable consumption practices: Linking organic food consumption to hedonic and eudaimonic well-being. *Appetite, 188*, Article 106633. https://doi.org/10.1016/j.appet.2023.106633

Ivanov, S., & Webster, C. (2017). Designing robot-friendly hospitality facilities. In *Proceedings of the scientific conference "tourism. Innovations. Strategies* (pp. 74–81). Bulgaria: Bourgas.

Ivanov, S., & Webster, C. (2018). Adoption of robots, artificial intelligence and service automation by travel, tourism and hospitality companies – A cost-benefit analysis. In V. Marinov, M. Vodenska, M. Assenova, & E. Dogramadjieva (Eds.), *Traditions and innovations in contemporary tourism* (pp. 190–203). Cambridge: Cambridge Scholars Publishing.

Jelodar, H., Wang, Y., Yuan, C., Feng, X., Jiang, X., Li, Y., & Zhao, L. (2019). Latent Dirichlet allocation (LDA) and topic modeling: Models, applications, a survey. *Multimedia Tools and Applications, 78*, 15169–15211. https://doi.org/10.1007/s11042-018-6894-4

Jin, E., & Eastin, M. (2023). Gender bias in virtual doctor interactions: Gender matching effects of chatbots and users on communication satisfactions and future intentions to use the chatbot. *International Journal of Human-Computer Interaction, 40*(23), 8246–8258. https://doi.org/10.1080/10447318.2023.2279402

Juraimi, S. A., Ze, W. L., & Pink, A. E. (2024). A pilot study examining the use of conversational AI-powered chatbots to understand consumer perceptions. *Appetite, 199*, Article 107469. https://doi.org/10.1016/j.appet.2024.107469

Kalla, D., Smith, N., Samaah, F., & Kuraku, S. (2023). Study and analysis of ChatGPT and its impact on different fields of study. *International Journal of Innovative Science and Research Technology, 8*(3). https://ssrn.com/abstract=4402499.

Kao, W.-K., & Huang, Y.-S.(S.) (2023). Service robots in full- and limited-service restaurants: Extending technology acceptance model. *Journal of Hospitality and Tourism Management, 54*, 10–21. https://doi.org/10.1016/j.jhtm.2022.11.006

Kim, H., Jung, S., & Ryu, G. (2020). A study on the restaurant recommendation service app based on AI chatbot using personalization information. *International Journal of Advanced Culture Technology, 8*(4), 263–270. https://doi.org/10.17703/IJACT.2020.8.4.263

Lähteenmäki, L. (2013). Claiming health in food products. *Food Quality and Preference, 27*(2), 196–201. https://doi.org/10.1016/j.foodqual.2012.03.006

Lessio, N., & Morris, A. (2020). Toward design archetypes for conversational agent personality. *2020 IEEE international conference on systems, man, and cybernetics*. Toronto, Canada: IEEE. https://doi.org/10.1109/SMC42975.2020.9283254

Leung, X. Y., & Wen, H. (2020). Chatbot usage in restaurant takeout orders: A comparison study of three ordering methods. *Journal of Hospitality and Tourism Management, 45*, 377–386. https://doi.org/10.1016/j.jhtm.2020.09.004

Liebrecht, C., Kamoen, N., & Aerts, C. (2023). Voice your opinion! young voters' usage and perceptions of a text-Based, voice-based and text-voice combined conversational agent voting advice application (CAVAA). In A. Følstad, T. Araujo, S. Papadopoulos, E. L.-C. Law, E. Luger, M. Goodwin, & P. B. Brandtzaeg (Eds.), *Chatbot research and design* (pp. 34–49). Springer International Publishing.

Liu, M., Yang, Y., Ren, Y., Jia, Y., Ma, H., Luo, J., Fang, S., Qi, M., & Zhang, L. (2024). What influences consumer AI chatbot use intention? An application of the extended technology acceptance model. *Journal of Hospitality and Tourism Technology, 15*(4), 667–689. https://doi.org/10.1108/JHTT-03-2023-0057

Luo, X., Tong, S., Fang, Z., & Qu, Z. (2019). Frontiers: Machines vs. humans: The impact of artificial intelligence chatbot disclosure on customer purchases. *Marketing Science, 38*(6), 937–947. https://doi.org/10.1287/mksc.2019.1192

Mai, R., & Hoffmann, S. (2015). How to combat the unhealthy = tasty intuition: The influencing role of health consciousness. *Journal of Public Policy and Marketing, 34*(1), 63–83. https://doi.org/10.1509/jppm.14.006

Maksimova, M., Solvak, M., & Krimmer, R. (2021). Data-driven personalized e-Government services: Literature review and case study. In N. Edelmann, C. Csáki, S. Hofmann, T. J. Lampoltshammer, L. Alcaide Muñoz, P. Parycek, G. Schwabe, & E. Tambouris (Eds.), *Electronic participation* (pp. 151–165). Springer International Publishing.

Mende, M., Scott, M. L., van Doorn, J., Grewal, D., & Shanks, I. (2019). Service robots rising: How humanoid robots influence service experiences and elicit compensatory consumer responses. *Journal of Marketing Research, 56*(4), 535–556. https://doi.org/10.1177/0022243718822827

Moussawi, S., & Benbunan-Fich, R. (2020). The effect of voice and humour on users' perceptions of personal intelligent agents. *Behaviour & Information Technology, 40*(15), 1603–1626. https://doi.org/10.1080/0144929X.2020.1772368

Muehlhoff, E., Bennett, A., & McMahon, D. (2013). *Milk and dairy products in human nutrition*. Rome, Italy: FAO.

Oğan, E. (2024). Artificial intelligence in restaurant businesses: A systematic review on service robots. *Worldwide Hospitality and Tourism Themes, 16*(2), 150–162. https://doi.org/10.1108/WHATT-03-2024-0058

Oh, Y. J., Zhang, J., Fang, M.-L., & Fukuoka, Y. (2021). A systematic review of artificial intelligence chatbots for promoting physical activity, healthy diet, and weight loss. *International Journal of Behavioral Nutrition and Physical Activity, 18*(1), 160. https://doi.org/10.1186/s12966-021-01224-6

Osei, B. A., Aflakpui, B. E., Adamtey, F. N. A., & Nutsugbodo, R. Y. (2024). Chatbot usage at fast food restaurants in Ghana: Exploring the perception-attitude-behaviour nexus. *Tourism and Hospitality Research*. https://doi.org/10.1177/14673584241303861

Papies, E. K. (2016). Goal priming as a situated intervention tool. *Current Opinion in Psychology, 12*, 12–16. https://doi.org/10.1016/j.copsyc.2016.04.008

Paul, J., Ueno, A., & Dennis, C. (2023). ChatGPT and consumers: Benefits, pitfalls and future research agenda. *International Journal of Consumer Studies, 47*(4), 1213–1225. https://doi.org/10.1111/ijcs.12928

Petty, R. E., & Cacioppo, J. T. (1986). *Communication and persuasion: Central and peripheral routes to attitude change*. New York, USA: Springer-Verlag.

Piao, M., Kim, J., Ryu, H., & Lee, H. (2020). Development and usability evaluation of a healthy lifestyle coaching chatbot using a habit formation model. *Healthcare Informatics Research, 26*(4), 255–264. https://doi.org/10.4258/hir.2020.26.4.255

Pliner, P., & Bell, R. (2009). A table for one: The pain and pleasure of eating alone. In H. L. Meiselman (Ed.), *Meals in science and practice: Interdisciplinary research and business applications* (pp. 169–189). Cambridge: Woodhead Publishing Ltd.

Prasetyo, P. K., Achananuparp, P., & Lim, E.-P. (2021). Foodbot: A goal-oriented just-in-time healthy eating interventions chatbot. *Proceedings of the 14th EAI international conference on pervasive computing technologies for healthcare*. New York, NY: ACM. https://doi.org/10.1145/3421937.3421960

Qiu, L., & Wan, X. (2024). Nature's beauty versus urban bustle: Chinese folk music influences food choices by inducing mental imagery of different scenes. *Appetite, 199*, Article 107507. https://doi.org/10.1016/j.appet.2024.107507

Rahmanti, A. R., Yang, H.-C., Bintoro, B. S., Nursetyo, A. A., Muhtar, M. S., Syed-Abdul, S., & Li, Y.-C. J. (2022). SlimMe, a chatbot with artificial empathy for personal weight management: System design and finding. *Frontiers in Nutrition, 9*, Article 870775. https://doi.org/10.3389/fnut.2022.870775

Roininen, K., Lähteenmäki, L., & Tuorila, H. (1999). Quantification of consumer attitudes to health and hedonic characteristics of foods. *Appetite, 33*(1), 71–88. https://doi.org/10.1006/appe.1999.0232

Rzepka, C., Berger, B., & Hess, T. (2022). Voice assistant vs. chatbot – Examining the fit between conversational agents' interaction modalities and information search tasks. *Information Systems Frontiers, 24*(3), 839–856. https://doi.org/10.1007/s10796-021-10226-5

Sam, S. J. I., & Jasim, K. M. (2023). Diving into the technology: A systematic literature review on strategic use of chatbots in hospitality service encounters. *Management Review Quarterly*. https://doi.org/10.1007/s11301-023-00388-8

Schwab, K. (2017). *The fourth industrial revolution*. New York: Crown Business.

Shumanov, M., & Johnson, L. (2021). Making conversations with chatbots more personalized. *Computers in Human Behavior, 117*, Article 106627. https://doi.org/10.1016/j.chb.2020.106627

Song, H., Kim, K., Kim, K., Ryu, J., & Suh, B. (2024). Exploring the impact of user-participated customization in experiencing chatbot failure. *Extended abstracts of the CHI conference on human factors in computing systems*. New York, NY: ACM. https://doi.org/10.1145/3613905.3651102

Suhaili, S. M., Salim, N., & Jambli, M. N. (2021). Service chatbots: A systematic review. *Expert Systems with Applications, 184*, Article 115461. https://doi.org/10.1016/j.eswa.2021.115461

Symmank, C., Mai, R., Hoffmann, S., Stok, F. M., Renner, B., Lien, N., & Rohm, H. (2017). Predictors of food decision making: A systematic interdisciplinary mapping (SIM) review. *Appetite, 110*, 25–35. https://doi.org/10.1016/j.appet.2016.11.023

Tian, Y., & Kamran, Q. (2021). A review of antecedents and effects of loyalty on food retailers toward sustainability. *Sustainability, 13*(23), Article 13419. https://doi.org/10.3390/su132313419

van der Goot, M. J., & Pilgrim, T. (2020). Exploring age differences in motivations for and acceptance of chatbot communication in a customer service context. In A. Følstad, T. Araujo, S. Papadopoulos, E. L.-C. Law, O.-C. Granmo, E. Luger, & P. B. Brandtzaeg (Eds.), *Chatbot research and design* (pp. 173–186). Springer International Publishing. https://doi.org/10.1007/978-3-030-39540-7_12.

Wald, R., Heijselaar, E., & Bosse, T. (2021). Make your own: The potential of chatbot customization for the development of user trust. *Adjunct proceedings of the 29th ACM conference on user modeling, adaptation and personalization*. New York, NY: ACM. https://doi.org/10.1145/3450614.3463600

Wan, X., Qiu, L., & Wang, C. (2022). A virtual reality-based study of color contrast to encourage more sustainable food choices. *Applied Psychology: Health and Well-being, 14*(2), 591–605. https://doi.org/10.1111/aphw.12321

Wang, C., Huang, J., & Wan, X. (2021). A cross-cultural study of beliefs about the influence of food sharing on interpersonal relationships and food choices. *Appetite, 161*, Article 105129. https://doi.org/10.1016/j.appet.2021.105129

Wang, C., & Wan, X. (2025). Eating contexts encourage sustainable food choices: The mediating role of the symbolic meanings of foods. *Appetite, 207*, Article 107896. https://doi.org/10.1016/j.appet.2025.107896

Wei, Y., Lu, W., Cheng, Q., Jiang, T., & Liu, S. (2022). How humans obtain information from AI: Categorizing user messages in human-AI collaborative conversations. *Information Processing & Management, 59*(2), Article 102838. https://doi.org/10.1016/j.ipm.2021.102838

Wilson, A. L., Buckley, E., Buckley, J. D., & Bogomolova, S. (2016). Nudging healthier food and beverage choices through salience and priming. Evidence from a systematic review. *Food Quality and Preference, 51*, 47–64. https://doi.org/10.1016/j.foodqual.2016.02.009

Wood, W. (2000). Attitude change: Persuasion and social influence. *Annual Review of Psychology, 51*(51), 539–570. https://doi.org/10.1146/annurev.psych.51.1.539

Yang, H., & Lee, H. (2019). Understanding user behavior of virtual personal assistant devices. *Information Systems and E-Business Management, 17*(1), 65–87. https://doi.org/10.1007/s10257-018-0375-1

Yoon, J., & Yu, H. (2022). Impact of customer experience on attitude and utilization intention of a restaurant-menu curation chatbot service. *Journal of Hospitality and Tourism Technology, 13*(3), 527–541. https://doi.org/10.1108/JHTT-03-2021-0089

Zeng, G., Chen, Z., & Zhong, S. (2024). "We Chinese just want meat!" An analysis of Chinese netizens' reactions to vegetarian advocacy. *Food Quality and Preference, 115*, Article 105128. https://doi.org/10.1016/j.foodqual.2024.105128

Zhai, F. Y., Du, S. F., Wang, Z. H., Zhang, J. G., Du, W. W., & Popkin, B. M. (2014). Dynamics of the Chinese diet and the role of urbanicity, 1991–2011. *Obesity Reviews, 15*, 16–26. https://doi.org/10.1111/obr.12124

Zhou, J., Chen, Z., Wan, D., Wen, B., Song, Y., Yu, J., Huang, Y., Ke, P., Bi, G., Peng, L., Yang, J., Xiao, X., Sabour, S., Zhang, X., Hou, W., Zhang, Y., Dong, Y., Wang, H., Tang, J., & Huang, M. (2024). CharacterGLM: Customizing social characters with large language models. *Proceedings of the 2024 conference on empirical methods in natural language processing: Industry track*. Miami, Florida, US: Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-industry.107